

File permissions in Linux

Project description

In this project, I examined and managed file permissions on a Linux system for a large organization's research team. Using Linux commands, I identified unauthorized permissions on files and directories and updated them to ensure only authorized users had appropriate access. This helps maintain the security and integrity of the organization's file system.

Check file and directory details

I used the following command to check the permissions of all files and directories in the `/home/researcher2/projects` directory:

```
ls -la
```

The output showed the following permissions:

```
drwxr-xr-x researcher2 research_team .
drwxr-xr-x researcher2 research_team ..
--w----- researcher2 research_team .project_x.txt
drwx--x--- researcher2 research_team drafts
-rw-rw-rw- researcher2 research_team project_k.txt
-rw-r----- researcher2 research_team project_m.txt
-rw-rw-r-- researcher2 research_team project_r.txt
-rw-rw-r-- researcher2 research_team project_t.txt
```

Describe the permissions string

Example: `project_k.txt` had the permission string `-rw-rw-rw-`

The 10-character string is broken down as follows:

Character	Meaning
-	This is a regular file
rw-	User: has read and write access
rw-	Group: has read and write access
rw-	Other: has read and write access (unauthorized)

Change file permissions

The organization does not allow others to have write access to any files. `project_k.txt` had write permission for others, so I removed it using:

```
chmod o-w project_k.txt
```

After running `ls -la project_k.txt`, the permission changed from `-rw-rw-rw-` to `-rw-rw-r--`, confirming others can no longer write to this file.

Change file permissions on a hidden file

The hidden file `.project_x.txt` had been archived and should not be writable by anyone. Only the user and group should be able to read it. I used the following command:

```
chmod u-w,g-w,g+r .project_x.txt
```

The permission changed from `--w-----` to `-r--r-----`, confirming no one has write access and both the user and group can read the file.

Change directory permissions

Only researcher2 should be allowed to access the drafts directory. The group had execute permission, which needed to be removed. I used:

```
chmod g-x drafts
```

The permission changed from `drwx--x---` to `drwx-----`, ensuring only researcher2 can access the directory.

Summary

In this project, I used Linux commands to examine and update file permissions for the research team's directory. I identified three permission issues: `project_k.txt` had unauthorized write access for others, `.project_x.txt` had incorrect write permissions, and the `drafts` directory allowed group execute access. Using `chmod` commands, I corrected all permissions to ensure the system follows the organization's security policy and only authorized users have appropriate access.

```
researcher2@3b92d709fd82:~$ pwd
/home/researcher2
researcher2@3b92d709fd82:~$ ls
projects
researcher2@3b92d709fd82:~$ cd projects
researcher2@3b92d709fd82:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Aug 24 17:48 .
drwxr-xr-x 3 researcher2 research_team 4096 Aug 24 17:50 ..
-rw--w---- 1 researcher2 research_team 46 Aug 24 17:48 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Aug 24 17:48 drafts
-rw-rw-rw- 1 researcher2 research_team 46 Aug 24 17:48 project_k.txt
-rw-r----- 1 researcher2 research_team 46 Aug 24 17:48 project_m.txt
-rw-rw-r-- 1 researcher2 research_team 46 Aug 24 17:48 project_r.txt
-rw-rw-r-- 1 researcher2 research_team 46 Aug 24 17:48 project_t.txt
researcher2@3b92d709fd82:~/projects$ chmod o-w project_k.txt
researcher2@3b92d709fd82:~/projects$ ls -la project_k.txt
-rw-rw-r-- 1 researcher2 research_team 46 Aug 24 17:48 project_k.txt
researcher2@3b92d709fd82:~/projects$ chmod u-w,g-w,g+r .project_x.txt
researcher2@3b92d709fd82:~/projects$ ls -la .project_x.txt
-r--r----- 1 researcher2 research_team 46 Aug 24 17:48 .project_x.txt
researcher2@3b92d709fd82:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Aug 24 17:48 .
drwxr-xr-x 3 researcher2 research_team 4096 Aug 24 17:50 ..
-r--r----- 1 researcher2 research_team 46 Aug 24 17:48 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Aug 24 17:48 drafts
-rw-rw-r-- 1 researcher2 research_team 46 Aug 24 17:48 project_k.txt
-rw-r----- 1 researcher2 research_team 46 Aug 24 17:48 project_m.txt
-rw-rw-r-- 1 researcher2 research_team 46 Aug 24 17:48 project_r.txt
-rw-rw-r-- 1 researcher2 research_team 46 Aug 24 17:48 project_t.txt
researcher2@3b92d709fd82:~/projects$ chmod g-x drafts
researcher2@3b92d709fd82:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Aug 24 17:48 .
drwxr-xr-x 3 researcher2 research_team 4096 Aug 24 17:50 ..
-r--r----- 1 researcher2 research_team 46 Aug 24 17:48 .project_x.txt
drwx----- 2 researcher2 research_team 4096 Aug 24 17:48 drafts
-rw-rw-r-- 1 researcher2 research_team 46 Aug 24 17:48 project_k.txt
-rw-r----- 1 researcher2 research_team 46 Aug 24 17:48 project_m.txt
-rw-rw-r-- 1 researcher2 research_team 46 Aug 24 17:48 project_r.txt
-rw-rw-r-- 1 researcher2 research_team 46 Aug 24 17:48 project_t.txt
researcher2@3b92d709fd82:~/projects$
```